

GHG MITIGATION TECHNICAL GUIDE

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Summary of Acronyms and Abbreviations

AV = autonomous vehicle

BAU = business as usual

BCI = compression ignition (biodiesel)

BEB = battery electric bus

BEV = battery electric vehicle

BRT = bus rapid transit

CFB = Carbon Free Boston

CI = compression ignition (diesel)

CTU = city, township, unorganized area

CUD = compact urban development

EIA = Energy Information Administration

FAF = freight analysis framework

HEV = hybrid electric vehicle

ICE = internal combustion engine

LRT = light rail transit

MSP = Minneapolis St. Paul 7-county metropolitan area

NHTSA = National Highway Transportation Safety Administration

NTD = national transit database

PHEV = plugin hybrid electric vehicle

PLDV = passenger light-duty vehicle

PMT = person (or passenger) miles travelled

RI = intercity (commuter) rail - Northstar

RTDM = regional travel demand model

SAV = shared autonomous vehicle

SAEV = shared, electric, and autonomous vehicle

SI = spark ignition (gasoline)

TVMT = ton vehicle miles travelled

VMT = vehicle miles travelled

VOTT = value of travel time (in dollars)

1. Business-as-usual (BAU) from 2018 to 2040 elements

1.1. Introduction...

Detail the key elements in **business-as-usual** in 2040 when quantifying carbon emissions

Ramaswami, Fang, and Tabory (2020) provide a framework for urban planning tools for greenhouse gas (GHG) mitigation as shown in Figure 1 as four key levers of action based on the framework developed in previous work by the **UN International Resource Panel** (IRP, 2018). Similar to the **3Rs** of waste management, the first lever represents a reduction of land use through compact urban development (CUD). Lever 2 focuses on technological solutions in a single infrastructure system (e.g., vehicle fuel efficiency and electrification). Lever 3 encompasses improvements from **synergistic** solutions between infrastructure systems. The final lever is policy to change individual and business behavior.

Lever 1: Compact urban development—land use strategies reducing demand for infrastructure services	Lever 2: Infrastructure technology strategies making demand and supply more efficient	Lever 4: Policy and behavior change strategies affecting behaviors of individuals (and businesses) using infrastructure services
	Lever 3: Cross-infrastructure strategies leveraging circular economy strategies enhanced by compact development	

Figure 1 Four levers for urban transportation and land use decarbonization

A combination of the above levers is necessary to reach decarbonization targets. Given the residency time of GHG emissions, the cumulative emissions over a forecast horizon (i.e., between today and 2050) means that the response time of a lever is a critical factor. Lever 1, CUD, requires changes in land use that will take many years (if not decades) to enact. It is found that transit and **car-sharing** become cost-effective beyond a density of 35 **dwelling units** per acre (Santasieri, 2014) and 5-10 dwelling units per acre (Millard-Ball, 2019), respectively. Using the **Census Bureau** measure of density being population per land area (U.S. Census Bureau, 2011), 22 of the 186 Cities, Townships, and Unorganized Areas (CTUs) in MSP meet the **transit threshold** (representing 43% of households) and 109 meet the lower **car-sharing threshold** (representing 94% of the population). These results mean that the majority of MSP households currently live in areas where individual vehicles will be a more effective solution than mass transit. As such, efforts to decarbonize the private transportation fleet will be a key strategy for these areas.

Many of the strategies categorized under levers 2 and 4 can be implemented within a shorter time horizon of perhaps 1-10 years. For example, the average private automobile is retired after 14 years, so ~~that~~ most households could replace their existing vehicle with an electric vehicle within the next 5-10 years. It should also be noted that there are **synergies** between these levers. Pricing policies categorized under lever 4 increase the cost of travel and may encourage changes in land development and residential choice in the long-term. Similarly, lever 3 focuses on **circular economy** principles that encourage development within the existing **urban footprint**.

Commented [1]: @jason.hawkins@mail.utoronto.ca
Does it make sense to cut this section since much of the material is now in the BAU guide?

Commented [2]: Yes, it can be cut now.

Commented [AJ3]: Is even a term like this something we should define for people trying to use the tool?

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Commented [AJ5]: Defined in list of acronyms, but maybe define here fist: City, Township, Unorganized area

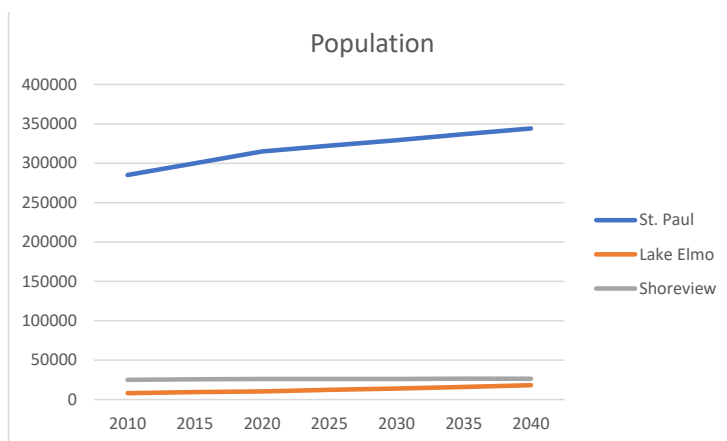
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Throughout the technical documentation, the CTU of St. Paul, Lake Elmo, and Shoreview are used as exemplars to simplify exposition.

1.2. Population Change

- Average growth across region is x%
- X% is happening in urban edge vs urban core



1.3. Economic Growth

- Increase in service economy/digital economy

1.4. Land Use Change & Urban Growth (Mike)

- Based on the land use comprehensive planning 2018 to 2040
- The ratio of single-family homes versus multi-family homes does not change, even for places with upzoning

BAU land use forecasts by the Metropolitan Council rely on a state-of-practice Cube Land model (Metropolitan Council, 2019a). Cube Land represents the location of households and establishments using a set of microeconomic models and is integrated with the regional transportation demand model. Subsidies and taxes, supply and demand restrictions, and planned land use changes derived from municipal comprehensive plans are among the inputs to the model. The inclusion of the comprehensive plans means that the BAU already anticipates the impacts of changes in zoning on real estate development and transportation Vehicle Miles Traveled (VMT) by mode.

1.5. Infrastructure sectors demand & supply (limited no more than one page for each of the following bullet)

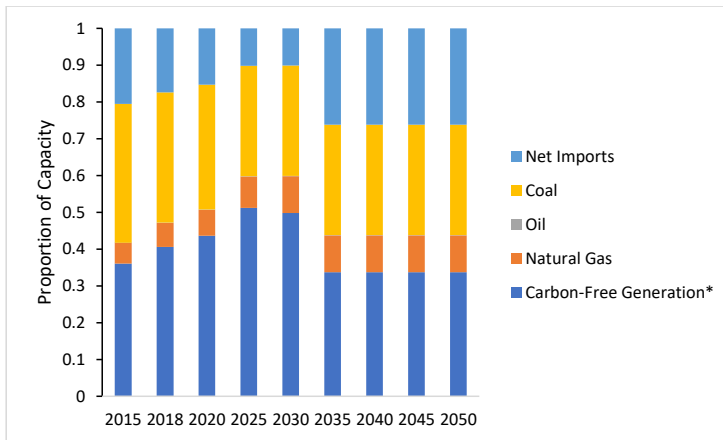
- 1.5.1 Energy demand & supply

Electricity supply is an important input to GHG intensity of the electricity grid, which is based on a weighted average of emissions factors derived from eGrid data for 2018 (U.S. EPA, 2019). Net imports are assumed to come from the neighboring states of North Dakota, South Dakota, Iowa, and Wisconsin. The average eGrid intensities are applied for these states. It is then assumed that changes in the GHG intensity of imports follow the same trend as that set out by Xcel Energy for Minnesota. The AR4-IPCC Panel 4 global warming potentials (as applied in eGrid 2018) are used throughout the tool and given in Table 1.

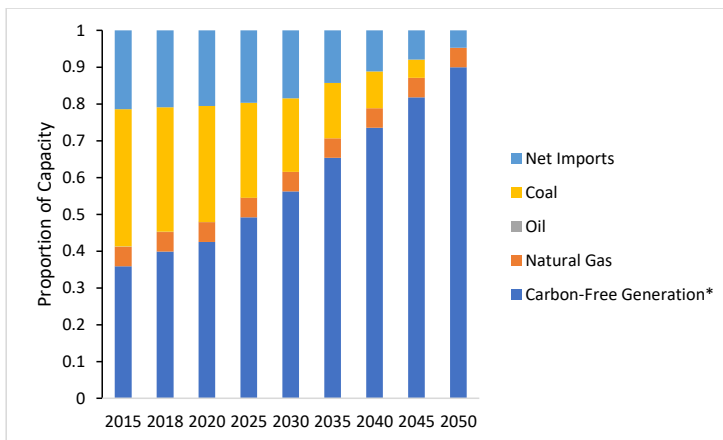
Table 1 Global Warming Potential Factors used in GHG Tool

Greenhouse Gas	Global Warming Potential
CO2	1
CH4	25
N2O	298

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(a) Reference Scenario



(b) Mitigation Scenario

Figure 2 Forecasted Electricity Generation Mix (*Carbon-Free Generation includes utility-scale solar, onshore wind, biomass, hydro-electric, and nuclear generators)

- Energy demand change due to efficiency improvement:
 - Xcel Energy proposed 2.5% annual electricity sales reduction and 1.4% annual natural gas sales reduction
- Incorporating the energy demand change due to climate change using the % of growth shown in a paper
 - 11% electricity use increase in the summertime due to climate change
- Grid level decarbonization and achieving carbon neutrality in 2050
 - Also used the base year's emission factor to evaluate the impact of grid decarbonization
 - 2040 the carbon emission factor for grid electricity is 128 lb CO₂/MWh

o District energy systems

- No expansion based on the past 10 years' record.
- Decarbonizing energy supply: 7% carbon reduction per year based on their carbon neutrality plan

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1.6. Transportation

The vehicle fleet can be decomposed into two segments: passenger and freight. Within the passenger transportation segment, the tool distinguishes between travel mode and powertrain.

1.6.1 Passenger Transportation

The dominant mode of passenger transportation in MSP is the private car (i.e., passenger light-duty vehicle). Powertrain types included in the tool for this mode are gasoline, diesel, hybrid electric-gasoline (HEV), plug-in hybrid electric-gasoline (PHEV), and battery electric (BEV). Public transit modes are categorized as urban bus, bus rapid transit (BRT), urban rail (LRT), and interurban rail (Northstar commuter rail). Bus modes have fuel options of biodiesel, hybrid electric-biodiesel, and battery electric. The LRT is already run by electricity and it is assumed that use of this technology will continue. The Northstar commuter rail currently uses biodiesel as an energy feedstock, but the option is available in the tool to switch it to an electric powertrain by 2050.

Commented [AJ15]: Simply bus routes in urban areas?

In the case of PHEV, there is a need to consider the fraction of VMT driven in electric and gasoline modes. It is assumed that the average travel distance in electric mode for a PHEV is 30 miles (Loveday, 2019) and that households only recharge vehicles at night. Using these assumptions, the fraction of time operating in gasoline mode can be estimated based on 2019 TBI data for household vehicle usage for each CTU. The regional model provides Person or Passenger Miles Traveled (PMT) totals for auto, transit, bike, and walk modes. PMT by mode is allocated to CTU using the standard 100-50-50 method given in the GHG Protocol for Cities (World Resources Institute et al., 2012). That is, 100% of PMT for trips that start and end in the same CTU is allocated to that CTU. Otherwise, 50% of PMT is allocated to each of the trip start CTU and trip end CTUs. A factor of 340 is used to convert daily PMT into annual PMT based on the recommendation of the Carbon Free Boston study (Porter et al., 2019).

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Average vehicle occupancy (AVO) is calculated for each CTU from the ratio of all automobile trips to automobile trips as driver in the 2019 TBI. These estimates are then validated against a region-wide AVO of 1.29 (Metropolitan Council, 2012).

Data for public transit are less readily available from the Regional Travel Demand Model (RTDM) compared with automobile travel. Forecasts are provided for total transit PMT, but no distinction is made between the various public transit modes (i.e., bus, BRT, LRT, and Northstar commuter rail). In addition, there are no estimates of transit vehicle turnover to the 2050 forecast year. For PMT by mode, detailed mode definitions are adapted from the 2019 TBI to estimate transit mode shares for each CTU. The same 100-50-50 heuristic is used as in the PMT estimation to obtain the factors shown in Table 1 below.

Table 2 Allocation of detailed transit modes among GHG tool transit modes for exemplar cities

CTU	Detailed Transit Mode	Detailed Proportion	GHG Tool Transit Mode Definition	Aggregated Proportion
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Lake Elmo	Express bus	1.0	Bus	1.0
Shoreview	Express bus	1.0	Bus	1.0
St. Paul	Local bus	0.31	Bus	0.52
St. Paul	Express bus	0.21		
St. Paul	BRT (i.e., A Line, C Line, Red Line)	0.27	BRT	0.27
St. Paul	Light rail (i.e., Blue Line, Green Line)	0.21	LRT	0.21
St. Paul	Northstar	0.0003	RI	0.0003

Transit **vehicle stock** and purchase forecasts begin from reported totals for the Metro Transit system in 2015 and 2020 by fuel type (Metro Transit, 2016, 2021a). Metro Transit purchased eight battery-electric buses (BEB) in 2020 and plans to have 130 BEB in-service by 2025. Vehicle forecasts beyond 2020 are based on anticipated PMT growth and an assumption that this growth is accommodated by additional vehicles rather than increased AVO. This assumption is deemed conservative and therefore provides a worst-case stock increase for the BAU scenario. Vehicle stocks and sales by propulsion technology are then allocated among the CTU based on PMT proportions. Bus fleet electrification is assumed to occur over the time horizon of the tool. Diesel buses are phased out by 2030 and replaced by a combination of BEB and HEV, with the HEV stock being negligible by 2050. This assumption is aligned with the plans of Metro Transit. While there have been vehicle range issues with the BEB fleet in MSP (Reilly, 2021), it is reasonable to assume that these issues will be addressed in the coming decades as the market grows. Including HEV in the short-term forecast also includes a contingency for longer routes. Ultimately, the benefits of BEB relative to diesel buses simply necessitate the transition. Recent research suggests that bus electrification could represent a health benefit (in terms of reducing premature death) as electrifying half the personal vehicle fleet of a North American city (Minet et al., 2021).

Transit vehicle retirements are based on 2019 data and useful life given in the National Transit Database (NTD) (FTA/NTD, 2021). Existing vehicle stocks are estimated by powertrain each year as the existing stock in the previous time step plus sales in the previous time step, less sales from two time steps prior (i.e., vehicles are replaced every 15 years).

The **fuel economy** values for public transit are taken from manufacturer specifications (O'Dea, 2018). These fuel economies are then forecast to 2050 using the **AEO** forecasts for passenger vehicles corresponding to each fuel type (i.e., diesel, hybrid, and battery electric) (EIA, 2021). **FHWA** estimates the bus AVO for MSP as 9.8 (FHWA, 2018), which we adjust based on additional data from the 2019 Metro Transit performance report (Metropolitan Council, 2019b). This report provides passengers per in-service hour for core local bus, suburban local bus, arterial BRT, LRT, and commuter rail. Factors are derived from these statistics relative to the aggregate regional passengers per in-service hour to obtain the AVO given in Table 2. It is assumed that **'core local bus'** is representative of service in St. Paul (and other urban CTUs) and **'suburban local bus'** is representative of service in Lake Elmo and Shoreview (and

other suburban CTUs). For validation, FTA (2010) gives an average seat occupancy of 30% for Metro Transit, which equates to an AVO of 16.8 for a 54 seat bus. The NTD is used for additional AVO imputation by mode using the reported ratio of passenger miles traveled to vehicle miles traveled for Metro Transit in 2015 and 2019 (FTA/NTD, 2021).

Table 3 Average vehicle occupancy (AVO) factors by transit service type

Transit service	Passengers per In-Service Hour	AVO
Core local bus	37.1	11.4
Suburban local bus	16.8	5.2
Arterial BRT	43.3	13.3
LRT	204.2	62.8
Regional total (bus only)	31.8	
Regional total (all transit)	26.4	
Commuter rail	249.2	76.7
Local bus (FHWA)		9.8 (given)
School bus		49.5 (assume 75% full on average (Chester and Horvath, 2009))

On-road vehicle fuel economy forecasts are taken from the [EIA Annual Energy Outlook \(AEO\)](#) for 2021, which forecasts fuel economy to 2050 (EIA, 2021). Base year values are provided for passenger vehicles in Table 3 and for transit vehicles in Table 4.

Table 4 Passenger vehicle fuel economy (common across CTU)

Powertrain Type	Fuel Economy (units as given) (US Department of Energy, 2020)
Gasoline	27.09 mpg
Diesel	31.66 mpg
HEV	45.48 mpg
PHEV (gasoline)	39.96 mpg
PHEV (electric)	2.72 mpkWh
BEV	3.13 mpkWh

Table 5 Transit vehicle fuel economy (common across CTU)

Vehicle/Powertrain Type	Fuel Economy (units as given) (FTA/NTD, 2021; O'Dea, 2018)
Biodiesel bus	4.82 mpg
HEV bus	5.84 mpg
BEV bus	0.42 mpkWh (taken as the national average)
Electric urban rail (LRT)	0.13 mpg
Biodiesel commuter rail (Northstar)	9.38 mpg
Electric commuter rail (Northstar)	0.10 mpkWh (taken as the national average)

1.6.2 Freight Transportation

The tool can consider changes in the fuel mix for the freight modes of **single-unit truck**, **combination-unit truck**, and rail. A review of the literature suggests that electrification of water and air modes is possible, but there are no literature values for their anticipated fuel efficiency. Other fuel types (e.g., biofuels) are available, but it was decided that including these fuels in the tool was outside the scope of analysis. Fuel economy assumptions are summarized in Table 5.

Table 6 Freight vehicle fuel economy (common across CTU)

Vehicle Class	Fuel Economy (units as given) (AFDC, 2015; Earl et al., 2018; Environmental Protection Agency et al., 2016; Garcia-Alvarez et al., 2013)
Single-unit diesel truck	98.0 tmpg
Single-unit BEV truck	2.34 tmpkWh
Combination-unit diesel truck	128.2 tmpg
Combination-unit BEV truck	4.3 tmpkWh
Diesel freight rail	477 tmpg
Electric freight rail	0.489 tmpkWh
Multi-modal diesel freight	Average of truck and rail modes
Waterborne marine fuel freight	647 tmpg
Air jet fuel freight	12.6 tmpg

The **FHWA** Freight Analysis Framework provides estimates for freight movements in the United States, with forecasts to 2045 (Bureau of Transportation Statistics, 2020). Freight movements are categorized

according to the following modes: truck, rail, water, air, multi-modal (a combination of truck and rail, as well as mail shipments), and pipeline. The database was filtered to include only **ton-miles** of freight moved within the MSP region, or moved domestically with an origin or destination in MSP. A similar convention is applied for assigning ton-vehicle miles traveled (**TVMT**) for freight travel as for VMT for passenger travel. That is, 100% of TVMT for trips with both the origin and destination in MSP and 50% of TVMT for trips with one of the origin or destination in MSP. These freight flows are only for long-distance trips and the FAF does not provide sufficient spatial resolution to assign among CTU within MSP. However, FHWA also provides a network database that provides a detailed representation of TVMT on roadways within MSP. The network database also includes estimates of short-distance freight VMT. The combination of these data sources is used to estimate truck TMVT for each CTU, including both short-distance and long-distance trips. A distinction is then made between **single-unit** (SU) and **combination-unit** (CU) TVMT. The network database provides **AADT** estimates for each of these truck types but not ton-miles. Since CU vehicles will generally carry a heavier load than SU vehicles, Alternative Fuels Data Center (**AFDC**) data are used to estimate a ratio of 2.25:1 for CU:SU loads. The combination of this ratio and AADT distributions gives estimates for TVMT by SU and CU trucks.

It is assumed that all waterborne freight passes through St. Paul as this is the only CTU with a major river port. Truck vehicle stocks are split between diesel (CI) and battery electric (BEV) based on the Market Acceptance of Advanced Automotive Technologies (**MA3T**) forecasted distribution for the passenger fleet. A factor of 2/3 is applied to the BEV stock proportion to account for the anticipated lower market penetration in **long-haul freight** (Giuliano et al., 2021; Porter et al., 2019).

1.6.3 Embodied Emissions for Transportation

Embodied emissions are an important, but often overlooked, component of transportation decarbonization. The emissions associated with transportation include, not only direct tailpipe emissions, but also the emissions resulting from the manufacturing and maintenance of vehicles and infrastructure. The main source of embodied emissions for the tool is an extensive analysis by Chester and Horvath (2009). The unit of analysis is vehicle lifecycle and all emissions are allocated in the vehicle purchase year. Chester and Horvath include emissions from vehicle manufacturing, vehicle maintenance, tire production, and fixed costs and insurance. They also provide estimates of the marginal GHG emissions associated with each vehicle due to roadway construction, roadway maintenance, parking, and a variety of other resources. Passenger vehicle classes from the study that are included in the tool are light-duty gasoline sedan, light-duty gasoline truck, light-duty diesel sedan, and light-duty diesel truck. 2019 **TBI** vehicle data are used to determine the mix of vehicle classes for each CTU. For public transit, **the average embodied** emissions are used for diesel bus, LRT values are based on results for Boston's Green Line, and values for commuter rail are based on results for Chicago. **Embodied emissions estimates** are also available for school bus and EV bus.

The Chester and Horvath study does not include all the vehicle classes included in the GHG tool. Secondary sources were explored for HEV, PHEV, and BEV passenger vehicles; HEV diesel bus; and EV school bus. Most studies provide GHG emissions on a **per veh-km** or **pass-km** basis rather than for the vehicle lifecycle. In addition, other studies do not include the breadth of factors considered by Chester and Horvath. Manufacturing and maintenance are identified as the most widely available lifecycle stages included in studies of embodied GHG emissions. For alternative fuel passenger vehicles (i.e., HEV, PHEV, and BEV), vehicle-lifecycle values for manufacturing are taken from the **REET model** (REET, 2020) as the summation of components, assembly disposal and recycling (ADR), batteries, and fluids. The

electricity mix in GREET is set to MRO, which is the electricity reliability organization containing MSP. The GREET estimates are adjusted to match those of Chester and Horvath by taking a ratio of the light-duty gasoline sedan values provided in both sources. Vehicle-lifecycle maintenance estimates are taken from Xiong et al. (2019) for PHEV and BEV, with HEV maintenance being taken as the value for PHEV adjusted by the ratio of their manufacturing estimates. For public transit, HEV diesel bus manufacturing and maintenance are approximated using the ratio of per pass-km values for HEV and diesel bus from Nordelöf et al. (2019) to adjust the diesel bus estimates of Chester and Horvath. EV school bus values are based on an adjustment of the EV bus estimates by the ratio of diesel school bus and diesel transit bus. Freight vehicle stocks and sales are not available for MSP, so it is not feasible to include their embodied emissions without strong assumptions or additional data collection.

Emissions factors for fossil fuels are provided in Table 7.

Table 7 Emissions factors by transportation class and powertrain

Transportation Class	Source	GHG Emissions Factor (US EPA, 2018)	Units
Passenger transportation	Gasoline	8.887	kg/gal
	Diesel	10.18	kg/gal
	Biodiesel	9.45	kg/gal
Freight transportation	Medium and heavy truck	0.202	kg/ton-mile
	Rail	0.0232	kg/ton-mile
	Watercraft	0.0602	kg/ton-mile
	Aircraft	1.320	kg/ton-mile
	Multimodal	0.113*	kg/ton-mile

* Average of truck and rail modes

1.6.4 Annual Energy Outlook

In addition to the inclusion of the strategies provided in the tool, there are global uncertainties in key energy markets (i.e., oil and gas) and macroeconomic conditions. To capture these effects, alternative future scenarios are adopted from the EIA Annual Energy Outlook. The AEO scenarios are incorporated into the transportation component of the tool as variations in fuel economy and VMT by mode. The following AEO scenarios are included in the tool (EIA, 2021):

- High oil price:** the price of Brent crude oil, in 2020 dollars, reaches \$173 per barrel by 2050 compared with \$95 per barrel in the reference case.
- Low oil price:** the price of Brent crude oil, in 2020 dollars, drops to \$48 per barrel by 2050 compared with \$95 per barrel in the reference case.
- High economic growth:** U.S. GDP grows by 2.6% per annum compared with 2.1% in the reference case.

Commented [AJ18]: Need definition in summary of acronyms and definitions? Midwest Reliability Organization?

4. **Low economic growth:** U.S. GDP grows by 1.6% **per annum** compared with 2.1% in the reference case.

The **AEO** analysis includes two additional scenarios representing high and low renewable cost cases. We do not consider these cases as we rely on Xcel Energy electricity grid forecasts in the tool and variation in renewable prices would affect these assumptions.

1.6.5 Transportation Costs

The tool includes estimates of transportation costs based on per mile estimates from the literature and AEO forecasts of future costs. These costs vary depending on a range of assumptions and are not uniformly available for all modes and **powertrain** alternatives included in the tool. The methods for determining costs are categorized into passenger transportation, freight transportation, and future transportation - summarized below.

Passenger Transportation

The primary source for passenger transportation costs is the American Automobile Association (AAA, 2020). Their cost estimate includes depreciation, finance, fuel, insurance, license and registration, and maintenance and repairs. This average value is then adjusted for each **powertrain** based on a secondary cost estimate from the **NREL FASTSim tool** (Brooker et al., 2015). For transit vehicles, estimates are taken from Quarles et al. (2020), who consider the total cost of conventional and autonomous buses with either diesel or battery-electric powertrains. These estimates are converted to per mile values using their assumptions for annual **VMT** (54,403 miles per bus) and years of life (12 years per bus). Their estimates include the cost of vehicle acquisition, operating costs, and driver costs. These bus numbers are then translated into estimates for **BRT**, LRT, and commuter rail using ratios derived from passenger-mile costs in the NTD. Costs per passenger-mile are translated into costs per vehicle-mile using average passengers per vehicle rates by transit mode. Finally, school bus estimates are taken from a New York State Education Department report (NYSED, 2020). A summary of per mile costs for passenger modes is provided in Table 8.

Table 8 Passenger transportation per mile costs

Mode/powertrain	Cost (2020\$ per mile)
PLDV-SI	0.62
PLDV-CI	0.62
PLDV-HEV	0.55
PLDV-PHEV	0.49
PLDV-BEV	0.55
Bus-BCI	6.66
Bus-HEV	6.57

Bus-BEV	6.47
BRT-BCI	9.17
BRT-HEV	9.05
BRT-BEV	8.91
LRT-EV	10.43
RI-BCI	10.13
RI-EV	9.86
BS-CI	5.50
BS-BEV	5.35

Freight Transportation

Freight transportation costs are not considered in the tool as the team was unable to find reliable estimates for enough modes. Cost estimates are provided for trucks in Table 9 for information purposes only (Earl et al., 2018; Murray and Glidewell, 2019).

Table 9 Freight transportation per mile costs

Mode/ powertrain	Cost (2018\$ per mile)
Class 7 truck-Diesel (in Midwest region)	1.59
Class 7 truck-BEV	1.96

Future Transportation

The cost of future transportation technologies, autonomous vehicles (**AVs**) and dynamic ride-hailing **DRS**, remains uncertain. This section outlines a set of best guesses at these costs based on a variety of sources.

DRS costs (assuming an electric powertrain) include electricity, vehicle maintenance, insurance and registration, charger infrastructure and maintenance, and vehicle and battery purchase. While these costs are uncertain, they are estimated to be in the range of 39 to 83 cents per vehicle-mile, with a mid-cost estimate of 58.7 cents per vehicle-mile (Loeb and Kockelman, 2019). For reference, the cost of a gasoline-powered shared vehicle fleet would be about 45.1 cents per vehicle-mile. The additional cost of EVs is due to a higher purchase cost, extra empty VMT reducing potential **revenue-miles**, and the costs of battery replacement and charging station construction. The most profitable application of shared, autonomous, and electronic vehicles (**SAEV**) is long-range trips with charging by fast-chargers.

Commented [AJ19]: Dynamic ride sharing?

It should be noted that the above estimates include the fact that existing gasoline vehicle infrastructure is far more developed than EV infrastructure. Analysis by Deloitte provides an economic case for DRS through a comparison of personal and shared conventional and autonomous vehicles (Corwin et al., 2015). In their analysis, DRS costs \$0.31 per mile compared with \$0.97 per mile for a personally owned conventional vehicle and \$0.46 per mile for a personally owned AV. Most of the cost savings come from a reduction in the time cost of travel (i.e., a more enjoyable experience) and a lower depreciation cost due to the higher utilization of shared vehicles. In another study, conventional vehicles and AVs were found to have similar per mile costs. However, conventional vehicle costs were anticipated to remain stable to 2035, while AV costs were forecast to drop by almost 60% by the same year to \$0.30 per mile (Johnson and Walker, 2016). The drop in price follows from a standard logistic S-curve technology adoption projection. As the technology is developed, the per mile cost of the mode will decrease. This decreased cost (or a portion thereof) will be passed onto the user through lower fares. In the case of transit, AVs will reduce operating costs through the elimination of drivers or their replacement with lower-wage attendants (Quarles et al., 2020). Current costs are given in Table 10.

Table 10 Future transportation per mile costs

Mode/powertrain	Cost (2020\$ per mile)
PLDV-SI	0.62
PLDV-CI	0.62
PLDV-HEV	0.55
PLDV-PHEV	0.49
PLDV-BEV	0.55
Bus-BCI	2.91
Bus-HEV	2.82
Bus-BEV	2.73
BRT-BCI	4.01
BRT-HEV	3.88
BRT-BEV	3.76
LRT-EV	4.45
RI-BCI	4.43
RI-EV	4.21
BS-CI	2.43
BS-BEV	2.28

DRS-HEV	0.58
DRS-PHEV	0.52
DRS-BEV	0.59

1.6.6 Transportation GHG Impact Calculation

The tool uses a single generic equation to determine GHG emissions for passenger transportation in most cases as follows

$$GHG\ impact(in\ mt\ of\ eqCO_2) = \frac{PMT}{AVO \times FPM} \times GF \times SF$$

where PMT is travel in passenger-miles, AVO is an mode-specific average vehicle occupancy, **FPM** is a fuel factor representing the consumption of fuel (in gallons or kWh) per mile of travel, and GF is a GHG factor per unit of consumed fuel. GF depends on the GHG intensity of the electricity grid. SF is a scenario-specific factor that adjusts the GHG impact of transportation (additional details are provided in each of the transportation sub-sections of the report). The GHG intensity of the grid is outlined in **section XX** above.

2. Private Electric Vehicle Adoption

2.1. Penetration of Electric Vehicles in Minneapolis-St. Paul Metropolitan Region

Vehicle electrification is widely recognized as a central pillar of transportation decarbonization (Sustainable Development Solutions Network, 2020). The effectiveness of this strategy relies on a combination of market uptake and decarbonization of the electricity grid. Both factors are considered in the tool. Focus is placed on the personal vehicle fleet but partial consideration is given to the on-road freight fleet.

2.2. GHG Impact Calculation

In the case of EVs, the **FPM** factor is measured in kWh of electricity consumed per mile and GF depends on the GHG intensity of the electricity grid. The following subsection outlines the method used to forecast EV market penetration.

2.3. EV Market Penetration

The Market Acceptance of Advanced Automotive Technologies (**MA3T**) model developed by Oak Ridge National Laboratory (ORNL) was used to estimate the 2018 private vehicle fleet composition and provide a baseline forecast of its composition in 2050 (Lin and Greene, 2013). CTU were aggregated so that the model includes the 44 CTU with the largest populations and the remaining CTU were allocated to their respective county among the seven counties in the MSP region (giving 51 regions). The model was then updated to use population totals and other characteristics calculated for these regions. Area types (i.e., *central city, suburban, or rural*) were based on United States Census Bureau definitions of *central cities* for the metropolitan area including Minneapolis, St. Paul, and Bloomington and assuming that non-

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community CTUs are *rural*. The MA3T model defines three driving patterns as modest (less than 8,656 miles per annum), average (8,656-16,068 miles per annum), and frequent (greater than 16,068 miles per annum). Weighted 2019 TBI data were used to estimate the share of each driving pattern class for each of the 51 regions.

It was found that the estimate for EV penetration by 2020 is high compared with what is observed in the county totals for that year. The model predicts 4,920 PHEV and BEV vehicles in 2015 and 97,091 such vehicles in 2020 for the seven-county region. For comparison, the Alternative Fuels Data Center (AFDC) only shows 4,740 registered vehicles in 2018 (AFDC, 2018) and MNDOT shows 17,146 registered electric vehicles in 2020 (MNDOT, 2020). A large 5-year growth rate is anticipated by the model between 2015 and 2020 of 383%. Assuming that this growth will occur in 2020-2025, that is it is lagged from the MA3T expectation by five years, gave reasonably good results by 2040 of about 23% of the vehicle market and 13% for BEV only (in line with the AEO and ZCAP reference cases). Please include a flowchart and key parameters here

In the case of PHEV, there is a need to consider the fraction of VMT driven in electric and gasoline modes. It is assumed that the average travel distance in electric mode for a PHEV is 30 miles (Plötz et al., 2020) and that households only recharge their vehicles at night. Using these assumptions, the fraction of time operating in gasoline mode can be estimated based on 2019 TBI data for household vehicle use for each CTU.

2.5. Precedents

Globally, EV sales are on the rise and now account for 4.2% of global vehicle sales (Irle, 2019). China and Germany lead in terms of total sales, but Norway is the leader as a proportion of the total vehicle market. The United States has seen significant growth in EV sales since 2010. The White House is proposing a \$174 billion investment in the EV market with the following targets by 2030: installation of 500,000 EV chargers, replacement of 50,000 diesel buses, and 20% electrification of the school bus fleet (U.S. Office of the Press Secretary, 2021).

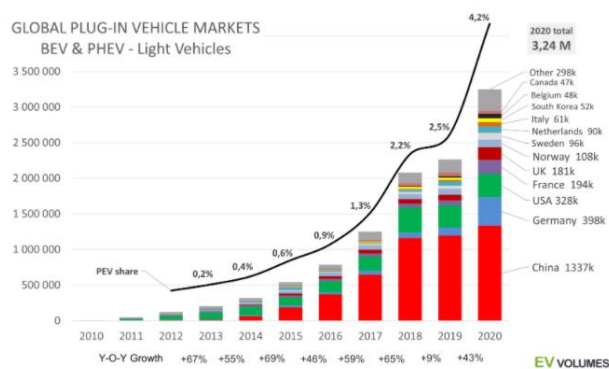


Figure 3 Growth in global plug-in EV market (Irle, 2019)

California is at the forefront of policy and adoption in the United States. Electric utility providers offer rebates to customers who drive EVs, financed by revenue from **low-carbon fuel standard (LCFS)** credits (California Public Utilities Commission, 2021). The state government also offers a rebate, totaling \$823 million since 2010. Finally, California has invested heavily in charging infrastructure. The proposed federal stimulus for EV adoption includes \$100 billion in new consumer rebates and \$15 billion for new charging stations (Shepardson, 2021).

Another precedent for the adoption of strong EV targets is the shift among automakers towards EV sales. Tesla is now valued at more than the combination of GM, Toyota, Volkswagen, and Ford (Boudette and Davenport, 2021). Partially in response, GM intends to sell only zero-emission vehicles by 2035. BMW, Ford, Honda, Volkswagen, and Volvo are bound under a deal with California to reach a standard of 51 mpg by 2026 (California Public Utilities Commission, 2021). This high fuel standard will likely require these automakers to increase their production of EVs and other zero-emission vehicles.

Minnesota is a member of the Midcontinent Transportation Electrification Collaborative (MTEC), a coalition of local governments, state transportation and environment agencies, and NGOs (Midcontinent Power Sector Collaborative, 2018). Within the state, Drive Electric Minnesota (DEMn) has, since 2010, been a platform for coordination of vehicle electrification between state and local governments, automakers and auto dealers, charging equipment providers, and NGOs. Minnesota has the goal of electrifying 20% of its personal light-duty fleet by 2030, with an equal share between PHEV and BEV (MNDOT et al., 2019). Xcel Energy, Great River Energy, and Otter Tail Power Company offer reduced electricity prices to Minnesota EV owners, equivalent to \$0.60 per gallon (Midcontinent Power Sector Collaborative, 2018). Xcel Energy plans to invest \$25 million over three years (2020-2022) on EV pilot programs and initiatives (Midcontinent Power Sector Collaborative, 2018). Thirteen states offer some form of tax incentive (sales tax reduction, income tax credit, or direct monetary incentive) on EV sales, but Minnesota currently does not offer such an incentive. The other plans currently in place by the state government focus on charging infrastructure, funded by the maximum 15% share of VW settlement funds (MNDOT et al., 2019).

2.6. Case Studies

Through efforts led by the Great Plains Institute (GPI) and Clean Energy Resource Teams (CERTs), 28 cities in Minnesota completed pilot studies. Some examples are as follows (GPI and CERTs, 2018):

- Red Wing installed a DC faster charger for \$22,094 in November 2018. The DCFC is free for use due to a cost-share between the municipality and local businesses.
- Three DCFC were installed along Lake Superior for \$10,000 to \$15,000 per site. The project is a seven-year pilot study formed through a collaboration between Arrowhead Cooperative, Great River Energy, the City of Grand Marais, and ZEF Energy (a DCFC service provider operating in Minnesota and Wisconsin).
- Edina installed a charging station in 2019 and has plans for additional stations.
- Fridley and Elk River purchased EVs to showcase their benefits to residents. In Elk River, the city purchased a PHEV Mitsubishi Outlander as a fleet vehicle.

2.7. Cost Estimation

A 2020 study by NREL found that the average cost of charging a BEV (including equipment) is \$0.14/kWh in Minnesota (Borlaug et al., 2020). In comparison to a conventional gasoline internal combustion engine (ICE) vehicle, the lifetime fuel cost savings are in the range of \$2,400 to \$11,000 and \$3,250 to \$9,500 for BEV and PHEV, respectively (assuming typical charging behavior and equipment costs in Minnesota).

The GHG abatement cost of a BEV subsidy is about 2017\$350-640 per tCO₂e (Gillingham and Stock, 2018). However, this cost does not consider spillover benefits in the future. These benefits arise from the long-run reduction in costs resulting from technological improvements, economies of scale, and shifts in investments towards low-emission electricity generation. In the case of EVs, the current marginal generation source is often fossil fuels, which means the current abatement cost will be much higher than anticipated with shifts in the electricity sector towards renewables and other low-emissions sources. Technological improvements resulted in battery costs dropping by 75% between 2009 and 2015 (U.S. DOE, 2015). Similarly, there is a *network effect* whereby additional EV sales make the installation of charging stations more profitable (Yu et al., 2016; Zhou and Li, 2018). The above factors make it challenging to determine the abatement cost of vehicle electrification but indicate that the cost will drop significantly relative to its current high value.

The 5th IPCC assessment predicts the BEV abatement cost to drop to \$0-100 per tCO₂e by 2030 in combination with low-carbon electricity (Sims R. et al., 2014). IEA estimates the abatement cost at -\$59 to \$25 for BEV and -\$113 to \$15 for HEV (at \$60 per barrel oil) (IEA, 2020).

2.8. Equity

The higher purchase price of EVs poses an equity challenge. This challenge would be exacerbated when combined with carbon pricing, which would increase the cost of travel for lower-income households who continue to drive lower-efficiency ICE vehicles. Rebates targeted at lower-income households could help to address this disparity.

2.9. Barriers and Enablers

Among the barriers to EV adoption are range anxiety, high upfront cost, and a lack of charging infrastructure. However, there are many promising developments in both policy and technology to advance the deployment of EVs. It is anticipated that EVs will reach price parity (without a subsidy) within the decade and battery prices will continue their current steady decline (BNEF, 2020b).

From an environmental perspective, lithium mining is seeing increasing scrutiny as the industry grows in response to demand from the EV and consumer technology markets. The current practice is either ore mining (as in Australia) or saline extraction (as in Chile and Argentina) (Volkswagen AG, 2021). A promising new source of lithium is geothermal water (Coren, 2021; Early, 2020). The lithium is extracted from a subsurface brine, which is heated by naturally occurring geothermal energy. The process is in the early stages of commercialization but shows promise, notably as a source of employment for engineers with expertise in subsurface resource extraction (i.e., oil and gas extracted through steam-assisted gravity drainage, or SAGD). Another challenge associated with increased production of EVs is their demand for scarce cobalt and nickel resources (Turcheniuk et al., 2018). Promising advances are being

made commercializing copper and iron cathodes and silicon anodes, which are more abundant resources and have a larger capacity (up to six times) to store lithium ions (i.e., more battery capacity).

3. Public Transit

The public transit component of the tool focuses on the effects of increasing transit ridership and average vehicle occupancy (AVO).

3.1. Transit Ridership and Average Vehicle Occupancy (AVO)

Transit ridership is affected by a myriad of factors: fare, network connectivity, land use, etc. Many of these effects are captured in the tool through **cross-elasticities** for automobile policy. However, the ability to directly increase transit ridership and average vehicle occupancy (AVO) is also provided in the tool. Increasing transit ridership in the tool assumes that additional vehicles are purchased to satisfy the increased VMT. Increasing AVO assumes that the same number of vehicles are dispatched for a fixed demand, but they are more efficiently routed to increase AVO (i.e., decrease VMT for a fixed PMT). These changes are incorporated into the embodied emissions calculations. Transit PMT grows over time in the BAU for most CTU, with the increased demand assumed to occur at a fixed AVO and translated into additional vehicles.

3.2. GHG Impact Calculation

The GHG impacts associated with changes in transit service are captured through direct changes in GHG emissions due to increased or decreased VMT, as well as changes in the indirect (embodied) emissions of transit through variations in the number of purchased transit vehicles. A key factor for the effectiveness of transit to reduce GHG emissions is the average vehicle occupancy (AVO). No specific policy is defined for this scenario. Rather, AVO is directly considered as a result of a generic set of policies and operational changes to service. When run full, a bus will reduce GHG emissions relative to a full private vehicle. However, at a national scale, a typical non-work private auto trip produces slightly lower GHG emissions than an average occupancy bus (on a per passenger-mile basis) - see Figure 4. Importantly, these comparisons are only true when comparing a conventional diesel bus with a conventional gasoline vehicle. Considering the full life cycle, GHG emissions for a battery-electric bus (BEB) are about 18% that of a gasoline-fueled sedan - compared with being roughly equivalent for a conventional diesel-fueled bus (Chester and Horvath, 2009). These comparisons are based on the US electricity grid circa 2006, which has seen a reduction in its carbon intensity over the intervening years. However, these older national averages are likely in line with the current grid in Minnesota, which is on the higher side of GHG emission production per kWh. Improvements will grow with the anticipated continued decarbonization of the electricity grid. For this reason, we do not consider alternative scenarios to BEB investment. It is simply assumed that bus purchases are BEB beginning in 2030 based on the necessity of this change and the global outlook (BNEF, 2020a; Sarabia, 2021; The White House, 2021; Wanek-Libman, 2020).

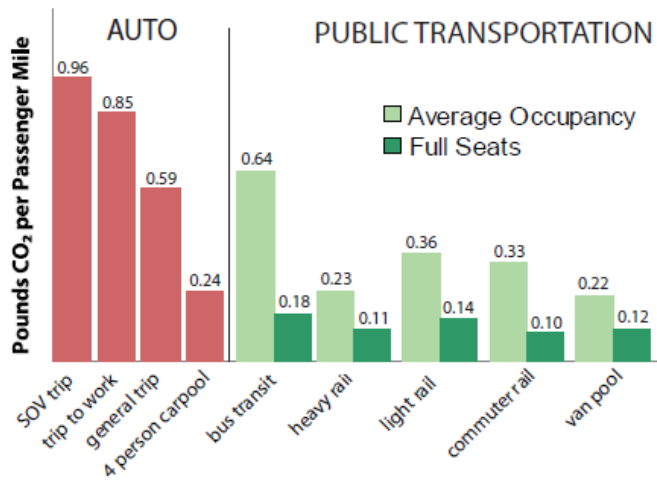


Figure 4 Summary of GHG emissions per PMT for average and full occupancy (US DOT, 2010)

3.3. Precedents

The future of transit demand remains unclear due to the effects of the COVID-19 pandemic. [Attitudinal](#) studies find that 75-80% of previous transit users intend to return when a vaccine is widely administered (Mashrur et al., 2020). APTA ridership data for the fourth quarter of 2020 reports a decline of 53% relative to 2019 (Dickens, 2020). Despite ridership uncertainty, many cities are investing in public transit to both invigorate their economy and build system capacity while ridership is low. The focus of many agencies is bus routes (based on more consistent demand over the past year than rail transit), improving service during off-peak periods, and introducing more dynamic service to better match supply with demand (Blumgart, 2021; TransLink, 2021).

While transit is generally associated with dense urban environments, it is not a precondition for an efficient and climate mitigating system. An evaluation of the GHG emissions for transit agencies in Ontario, Canada found that small agencies and transit in suburban communities can make a difference (Jami et al., 2021). Translating performance into a letter score, several agencies with 100 vehicles or fewer received grades of A- or better (including one agency with only 14 vehicles).

3.4. Case Studies

Before the emergence of the COVID-19 pandemic, Metro Transit put out the [Network Next](#) plan to improve bus service in MSP, with the major component being six new BRT routes (Metro Transit, 2021b). Work began on the Metro D Line in April 2021. A similar BRT expansion in Calgary, Canada (also a mid-size northern city) called MAX was associated with a 3.5% increase in ridership on their system in 2018 relative to pre-implementation in 2017 (Vaessen, 2019).

Indianapolis used new income tax revenue to finance a new BRT line, with plans for two additional lines and a total system extent of 97 stations over 62 miles (Renn, 2019). Similar to the MSP BRT network,

Indianapolis provides **off-board fare collection**. The total project cost is estimated at \$500 million, with a portion of this cost to be covered by federal sources. Currently, about 10% of Indianapolis residents do not own a car and rely on public transit.

Columbus provided free bus passes to downtown workers through a program called C-Pass (Renn, 2019). The program is funded through a transit improvement fund levied from local property owners. Following the introduction of the program, transit ridership rose from 5% to about 10-14%. The program includes an integrated fare structure between public transit and private taxicabs, partially funded by a \$50 million grant from the USDOT Smart City Challenge.

3.5. Cost Estimation

The **marginal abatement cost** for bus electrification is estimated at about -\$42-4236 per tCO₂e (Massoli, 2020; Stanton et al., 2019). This negative value means that there is a net monetary benefit associated with bus electrification. Note that this estimate was developed for Massachusetts and the GHG reduction is a modest 0.1 million tCO₂e because of the low transit mode share. For reference, the same study gives an abatement cost of \$50 per tCO₂e for passenger vehicle electrification but a total abatement of 5.7 tCO₂e. These estimates are taken from the Carbon Free Boston (CFB) study, which only considers project implementation costs (Porter et al., 2019). Inclusion of cost savings from improved health brought about by the elimination of particulate matter from diesel buses, among other societal benefits, would have a significant effect on the **LCCA**. IEA (2020) estimates the abatement cost at -\$74 to \$254 for BEB (at \$60 per barrel oil). Their average abatement cost is -\$7.10 for BEB or \$11.47 for EV bus (e.g., running on an **overhead catenary line**).

3.6. Equity

Equity plays an important role in weighing the benefits of transit investment. For many people, frequent and connected public transit is a vital service. However, from the perspective of GHG mitigation, the optimizing goal should be maximizing AVO.

3.7. Barriers and Enablers

A challenge to bus electrification in MSP has been that BEBs are deployed on longer routes and require frequent charging. It is anticipated that improvements in battery and BEB technology due to their widespread adoption will address this concern by 2030.

4. Land Use Change

4.1. Land Use Change in Minneapolis-St. Paul Metropolitan Region

Transportation policy cannot be separated from land use policy. Segregated and low-density land use patterns inevitably lead to reliance on the automobile, increased VMT, and a decreased efficacy of shared travel. However, altering land use policy faces significant resistance and does not immediately lead to changes in land use as construction takes time to catch up with zoning.

4.2. GHG Impact Calculation

Land use **elasticities** are applied as additional factors (SF in the generic transportation equation) to determine the effect of land use change on VMT and GHG emissions from transportation.

Land use factors that affect transportation demand are typically characterized according to the 5Ds of population and employment: **d**ensity, **d**iversity of land uses (i.e., residential, commercial, industrial), **d**esign (e.g., the proportion of 4-way intersections), **d**istance to transit, and **d**estination accessibility. **Elasticities** are available for all these features. However, there are a variety of factors to consider when assessing the effect of the 5Ds on transportation demand.

First, these features are **highly correlated**, and caution is required applying elasticities to avoid overstating impacts.

Second, the measures of each of the 5Ds must be correctly specified. For example, population density can be measured as a function of total land area, developed land area, or some other subset of the total land area. As a strategy to avoid overstating the effect of the 5Ds, a hard limit is set of 25% reduction in VMT based on previous findings and assumptions made in similar studies (Bento et al., 2005; Ramaswami et al., 2012; TRB, 2009). This limit approximates a transition in urban structure from Atlanta, GA to Boston, MA, which is also in-line with the earlier findings of Newman and Kenworthy (1989) regarding gasoline consumption (a surrogate measure of VMT) as a function of urban density. Stevens (2017) found that density and accessibility elasticities tend to double after controlling for self-selection. One reason given for this effect is that households are being forced into lower density, more auto-oriented neighborhoods than they desire, due to the high dwelling prices in more central and accessible neighborhoods. Given the opportunity, many households would choose to travel less (Chatman, 2009).

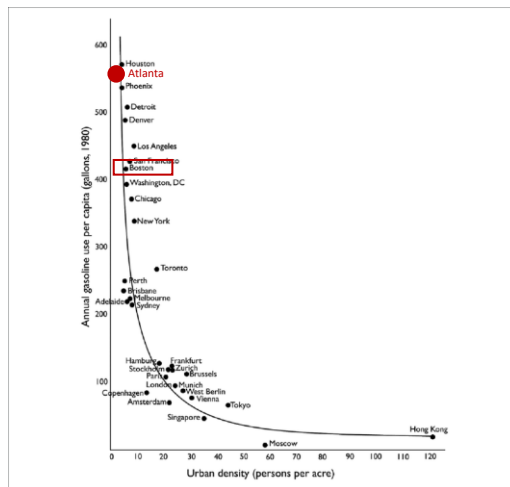


Figure 5 Annual gasoline use as a function of density across global cities (Newman and Kenworthy, 1989)

Elasticities are taken from a meta-analysis by Ewing and Cervero (2010) for each of automobile, walking and cycling, and transit trips (see Table 11). A more recent meta-analysis provides additional values for the automobile mode (Stevens, 2017). These values are used to give an elasticity range. In addition to the below values, an overall elasticity of -0.22 is used for the combined effect of the 5Ds to give the user the option to consider a combination of land use changes.

Population (employment) is calculated as the total population (jobs) in the CTU divided by the area of developed land. This definition allows for the simple comparison of alternative land development scenarios. Developed land includes both impervious surfaces and greenery such as grass and trees. These cover types are explored under the green infrastructure scenarios.

The user may specify changes in any (or all) of the 5Ds. However, the illustrative scenarios developed with the tool use a composite elasticity measured using changes in population density. This composite measure accounts for the fact that changes in one dimension of the 5Ds is often concomitant with changes in the other dimensions. The isolated population density elasticity will give an expected change in VMT for a change in density, holding constant the mix of land uses, average distance to transit, destination accessibility, and network connectivity.

$$\text{Scenario VMT} = 2040 \text{ BAU VMT} \times (1 + \text{elasticity} \times \text{percent change in density})$$

Please define each of the terms in equation and where they came from in your document

How is elasticity computed? Please show equation.

What is units of scenario VMT (daily?)

Please clarify units of density (population per unit urban area?)

Describe which elasticity was used when

Commented [AJ26]: Associated?

What additional land use changes can be modeled in the tool

Vehicle-fuel technology changes

Ridesharing

Behavioral/economic levers

Table 11 Summary of land use elasticities for VMT (numbers in parentheses give an indication of the range of elasticities found in the literature) *Did you use all of these elasticities in the model?*

Mode	Population Density	Employment Density	Diversity (Land Use Entropy)	Design (% 4-Way Intersections)	Destination Accessibility	Distance to Transit	Composite Effect of 5Ds
Automobile	-0.04 (-0.04 to -0.22)	0 (0.00 to -0.07)	-0.09 (-0.02 to -0.10)	-0.12 (-0.06 to -0.12)	-0.20 ¹	-0.05 (-0.05 to -0.08)	-0.22
Walking and cycling	0.07	0.04	0.15	-0.06	0.15 ²	0.15	0.33
Transit	0.07	0.01	0.12	0.29	³	0.29	0.62

¹ Measured as distance to CBD or jobs reachable within a given travel time

² Measured as jobs located within 1 mile of the home

³ Not provided in the Ewing and Cervero study

Commented [AJ27]: Central Business District?

Several of the above land use variables are summarized for representative CTU (Lake Elmo, Shoreview, and St. Paul) in Tables 12 to 15. Density is often measured as population per land area (Cohen, 2015). However, in CTUs such as Lake Elmo, there may be large areas that were undeveloped in 2015. There may also be areas that are parkland that cannot be converted to residential use. These areas should not be considered in the population density calculation. As such, density is measured using only the developed land area in each year. With these considerations, the increase in population density in Lake Elmo is 76%. Note that the percentage of residential land zoned for single-family detached dwellings is 97% for Lake Elmo in 2018 but is forecast to decrease to 93% by 2040.

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Table 12 Population and employment density by CTU

CTU	2018	2020	2030	2040
Population Density (per acre urban area)				
Lake Elmo	1.67	1.82	2.58	2.94
Shoreview	4.43	4.40	4.39	4.41
St. Paul	10.02	10.06	10.60	11.18
Employment Density (per acre urban area)				
Lake Elmo	0.47	0.48	0.48	0.50
Shoreview	2.03	2.14	2.28	2.39
St. Paul	6.10	6.47	6.86	7.24

Land use diversity is calculated using an **entropy measure** and the forecasted land uses. The “mixed-use” categories are assumed to contain two land uses for the purpose of calculating the **entropy term**.

$$entropy = \sum_i LU_i \ln(LU_i) \quad \text{eq. 1}$$

where LU_i is the proportion of land devoted to land use type i . **Entropy** is anticipated to increase in the more outlying CTUs, such as Lake Elmo and Shoreview, than in the central city of St. Paul.

Table 13 Current and forecasted land use entropy by CTU

CTU	2015	2020	2025	2030	2035	2040	2045	2050
Lake Elmo	0.44	0.46	0.48	0.50	0.51	0.53	0.55	0.57
Shoreview	0.41	0.42	0.43	0.44	0.46	0.48	0.50	0.53
St. Paul	0.62	0.61	0.59	0.58	0.59	0.61	0.63	0.64

OpenStreetMap street network data were used to calculate the distribution of intersection configurations in each CTU (OpenStreetMap Contributors, 2017). Service roads, parking lots, and several other features that are coded as streets in the data were removed from the network.

Table 14 Distribution of intersection configurations by CTU

CTU	% 3-way	% 4-way	% Other
Lake Elmo	75.9%	21.8%	2.2%
Shoreview	78.1%	20.6%	1.3%
St. Paul	42.9%	55.6%	1.5%

For each CTU, transit distance to each residential building was calculated using **OpenStreetMap** and **GTFS** data for MSP. Summary statistics are provided in Table 15 for the exemplar CTUs. As expected, St. Paul has the lowest mean distance to a transit stop and the smallest standard deviation.

Table 15 Summary statistics for distance to nearest transit stop (bus or rail) by CTU

CTU	Min	Max	Mean	Median	Std. Dev.	Q1	Q3
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Lake Elmo	185.9	4956.2	2269.1	1039.2	1956.9	457.9	4742.7
Shoreview	145.2	7818.0	1032.9	704.9	1098.4	533.1	992.9
St. Paul	23.7	2764.7	619.2	551.8	367.7	332.5	866.0

4.3. Precedents

The Minneapolis 2040 plan includes a series of changes to zoning to encourage multi-unit dwellings in that city (City of Minneapolis, 2019). The plan calls for additional density along public transit corridors and provides an allowance for up to three dwellings on a single lot in the most outlying neighborhoods. There are several caveats to interpreting this policy change as a precedent for other municipalities in MSP. Minneapolis is the largest city in the region and its high level of existing development means it has minimal land available for new developments. The higher price of land in Minneapolis, relative to other municipalities in MSP, provides an additional incentive for upzoning. It must also be borne in mind that this is an ongoing change and its contribution towards realizing increased density remains unclear.

A study by MAPC in Massachusetts used a combination of the VisionEval transportation model and the UrbanSim land use model to examine land use and pricing scenarios (Harmon, 2021). They found a 5.2% difference in VMT between smart growth and sprawl scenarios, equivalent to 2.3 million vehicle-miles traveled per day in that region.

4.4. Case Studies

Los Angeles is often cited as an exemplar low-density auto-oriented city. However, the city has invested heavily in rail transit over the last 20 years and provides a case study of transitioning an auto-oriented area to include transit-oriented development (TOD) along major transit corridors. The finding in Los Angeles is that the effectiveness of TOD depends on the objective (Schuetz et al., 2018). In the case of employment growth, benefits tend to be slow in areas that already serve as employment centers. Benefits tend to be more immediate and larger in magnitude when development occurs in greenfield areas, which provide a faster path to denser and more mixed residential development.

4.5. Cost Estimation

It is not a simple task to anticipate the GHG mitigation and implementation costs associated with changes to urban form owing to variability in its characteristics and its large scope. Whether development occurs in greenfield (i.e., previously undeveloped land) or brownfield (i.e., redevelopment of previously developed land) areas will influence whether there is a loss of carbon sinks. Costs will depend on the price of land and construction, as well as the need for new infrastructure. An element of the economic cost of urban form that has been quantified is the cost of restrictive zoning. A study of U.S. metropolitan areas estimates that restrictive zoning reduced economic growth by 36 percent between 1964 and 2009 (Hsieh and Moretti, 2019). This gap is attributed to restrictions placed on the ability of workers to access productive markets. While the cost of upzoning would be borne by a variety of stakeholders - cities, developers, landowners - there does appear to be a monetary benefit and the decarbonization benefits are numerous for both transportation and building energy.

4.6. Equity, Barriers, and Enablers

Research suggests that restrictive land use policies tend to increase both economic and racial segregation (Resseger, 2013; Rothwell and Massey, 2010). Pendall et al. (2006) found that restrictive zoning policy is most prevalent in Northeast and Midwest suburbs. Such policies tend to concentrate low and high-income households into systematically different communities because they segregate the land available for subsidized and multi-unit housing. In Boston, the difference in Black and Hispanic composition between adjacent blocks zoned for multi-family and single-family dwellings varied by 3.4 percentage points and 5.8 percentage points, respectively (Resseger, 2013).

One of the major impediments to the adoption of upzoning in the United States is the perception of opposition by the public. This issue is linked with equity considerations. Recent research examining the minutes from zoning board meetings found that existing longtime homeowners are the most likely attendees of these meetings (Einstein et al., 2019). Equity concerns arise from the fact that potential residents of the higher density developments made possible by upzoning are more likely to be renters and from minority groups. By their under attendance at these meetings, potential residents are unable to voice their support for a zoning change and policy makers may receive a biased impression of public opinion. It is suggested that through simple measures to more widely disseminate meeting announcements, local governments can obtain more representative feedback, which includes renters and potential residents of multi-unit dwellings and not just abutting property owners.

Commented [AJ29]: Better way to word this?
"Underrepresented residents who are unable..." Maybe mention 'Procedural Justice'

5. Autonomous Vehicles

Vehicle automation has the benefits of increasing driving **cycle efficiency** (i.e., fuel efficiency) and safety (Kockelman and Li, 2016). However, it is also anticipated that AVs will increase total VMT owing to an improved traveler experience by removing the driving task.

Given that this travel mode is not operational, implementation is based on a combination of literature sources and reasonable assumptions based on the research experience of the team.

The range of impacts on energy requirements (roughly translatable to GHG emissions) are summarized in Table 16. Energy reductions are anticipated to be in the range of 11% to 55% depending on whether the **powertrain** is electrified.

Table 16 Summary of energy impacts from vehicle automation and associated technologies (adapted from (Lee and Kockelman, 2019))

Category	Impacts	Description	Energy Impacts
Travel Demand Changes	Better route choices	Route choice based on real-time traffic data from connected environment	-5% to -20%
	New trips from under-served populations	Motorized trips by limited drivers & non-drivers	+10% to +14%

	Shared automated vehicles - empty driving	SAVs traveling to new passengers empty	+6% to +18%
	More long-distance travel	Longer distance travel caused by lower driving task of AVs	+7% to +30%
Driving Details	Faster travel from improved driving skills	Fast & throughput-efficient driving cycle	-10% to -20%
	Smoother driving cycle	More fuel-efficient driving cycle	+4% to +15%
	Computer & sensor power demands	Energy for sensors, on-board computing, vehicle control, & navigation	-2% to -19%
Multi-Vehicle Operations	V2V & platooning	Vehicle-to-vehicle connectivity & platooning	-5% to -12%
	Shared AVs & ride-sharing	Fuel savings from vehicle right-sizing & dynamic ride-sharing (DRS)	-6% to -30%
	V2I & smart intersections	Vehicle-to-infrastructure connectivity & smart intersections	-30% to -70%
Powertrain Energy Source	Plug-in electric & hybrid electric vehicles	Powertrain shifts from gasoline to electricity	

Commented [AJ30]: Smoother?

5.2. GHG Impact Calculation

It is assumed that VMT for AV trips is 20% higher than human vehicle (HV) trips (Barth and Boriboonsomsin, 2009; Harper et al., 2016; Huang et al., 2020; Zhao and Kockelman, 2018). AV vehicles may be HEV, PHEV, or BEV in the tool. Further, it is assumed that the drive cycle is more efficient for

these vehicles - measured as a reduction in GHG of 15% (Taiebat et al., 2019). The results of a simulation study of widescale AV adoption in the Texas Triangle are used as inputs to the tool for transit impacts (Huang et al., 2020). In scenario analysis, 45.6% of trips were made by AV and 17.0% of trips by SAV, resulting in a reduction in rail and bus VMT of 71.1% and 66.1%, respectively. An approximation is made that a 1% increase in AV market penetration induces reductions of 1.13% and 1.05% in rail and bus VMT, respectively. AV purchases are assumed to replace the existing stock in proportion to their VMT and 1/3 of the stock is replaced in each of the five-year time steps of the tool. A Bass diffusion model is used to interpolate AV market share between the implementation year of 2025 and horizon year of 2050 (Lavasani et al., 2016). The mode-specific effect of AVs can be introduced to the GHG emissions impact equation through the additional of a VMT factor. These factors are based on results for vehicle trips under 50 miles from a study conducted in Texas (see Table 17).

Table 17 Simulation results for the introduction of AVs in the Texas Triangle in 2040 (Huang et al., 2020)

	Private Car	Urban Rail	Bus
Change (wrt non-AV)	+46.7%	-77.1%	-49.8%

Commented [AJ31]: It's unclear what this acronym stands for

These results are highly uncertain. The effect of AVs on VMT depends upon a variety of factors, including the value of travel time (VOTT) and the cost of parking. Zhao and Kockelman (2018) found that a mix of AVs and SAVs would increase VMT in Austin, Texas by 18% to 41% for value of travel time (VOTT) reductions of 25% to 75% relative to human driven vehicles (HV).

5.3. Precedents

There are no large scale applications of AVs. MNDOT established an Office of Connected and Automated Vehicles in 2018, which supports the Governor’s advisory council with policy, planning, research, testing, and engineering analysis (MN DOT, 2019). The Minnesota CAV Challenge received 83 ideas to test CAV technology in the state, with 14 of these projects in the planning stage. A CAV corridor is envisioned along Highway 52 between Rochester and St. Paul.

5.4. Case Studies

There are no commercial AV systems in operations as of 2021, so case studies are taken from simulation studies. A series of studies by Kockelman and colleagues at the University of Texas at Austin found that AVs will increase VMT but may decrease net energy use by 11% to 55% (Fagnant and Kockelman, 2014; J Lee and Kockelman, 2019; Liu et al., 2017). The energy reduction is driven by a transition to BEVs and PHEVs, the improved driving cycle efficiency of AVs relative to human-driven vehicles, and better route choices if vehicles can communicate with each other.

In a similar spirit, Wadud et al. (2016) find that the impacts of AV may range from a halving to a doubling of GHG emissions relative to the current state. This variation is driven by whether AVs are electric and assumptions about the effect of AVs on total VMT, vis-à-vis reduced implicit travel time costs encouraging more travel.

5.5. Cost Estimation

Vehicle automation will likely increase the initial purchase cost but reduce fuel consumption and provide a more enjoyable trip (reducing the implicit travel time cost). These costs will vary by income group depending on the value of their current vehicle and their VOTT. A current estimate is that automation will increase the total cost of ownership by roughly \$965 per year (+7.9%) for the median household (Wadud, 2017). However, this cost ranges from \$1,916 (+29.6%) for the lowest income quintile to a savings of \$4,873 per year (-14.2%) for the highest 1% of households. The largest cost component is the sunk cost of the existing vehicle stock and the largest benefit is the productive use of travel time. Finally, these are early adopter costs, and most costs will certainly drop as the technology and manufacturing process develops and grows.

5.6. Equity

The above cost analysis suggests that lower-income households will experience a disproportionately negative outcome from AV ownership. The major source of differences among income groups is due to the larger productive time value of higher-income households, who are also more able to perform their work while traveling. Note, the relationship is likely flipped for DRS because lower-income households are more reliant on public transit. In the context of DRS, lower-income households will benefit from improved service relative to infrequent bus service.

5.7. Barriers/Enablers

There have been several draft bills in both the federal house and senate, but no legislation has been passed into law to date (National Conference of State Legislature, 2020). NHTSA has produced successive policy frameworks for managing AV adoption, but there are currently no federal laws or regulations governing the technology. The latest NHTSA report, AV 4.0, was released in January of 2020. Federal guidelines recommend that states allow USDOT to regulate safety design and performance aspects of AVs. State responsibilities are licensing, traffic laws, inspections, and insurance (Loftus-Otway et al., 2018).

At the state level, Minnesota Governor Mark Dayton issued Executive Order 18-04 in March of 2018, establishing an advisory council for the development of autonomous driving technologies (National Conference of State Legislature, 2020). The state adopted a law permitting connected truck platoons following the approval of the Departments of Public Safety and Transportation. A major barrier to the deployment of AVs in Minnesota is the severe winter weather conditions. MnDOT is working with private industry to perform low-speed shuttle tests in snow, ice, and white-out conditions (MN DOT, 2019).

6. Dynamic Ridesharing (DRS)

Dynamic ridesharing (DRS) is a system of shared autonomous vehicles (SAV) that are operated as a fleet. DRS is anticipated to reduce the need for vehicle ownership and the total number of vehicles on the road as a single vehicle can be shared by multiple users. A co-benefit of DRS, which is generally assumed to use AVs, is a significant reduction in parking - of the order of 80-90% (Fagnant and Kockelman, 2015; Kondor et al., 2020). However, it was found that most of the parking area anticipated to be freed up by

DRS is in residential areas, while the most expensive land is located in central business districts (Kondor et al., 2020).

6.2. GHG Impact Calculation

In a simulation study, Yan et al. (2020), considered three levels of DRS adoption in MSP: 2% of trips in the 7-county region, 5% of trips in the 7-county region, and 20% of trips in the twin cities of Minneapolis and St. Paul. Using 2% of trips, an SAV serving 5 travelers would have an average VMT of 170 miles per day. The VMT without a traveler (eVMT) would be 11.7% of the total and the SAV would make about 19 trips per day. Travelers have an average wait time of 4.0 minutes. Similar results were obtained for the 5% and 20% samples, with a higher VMT per SAV and a similar average wait time. These results are used to build a DRS capability into the tool. It is assumed that each DRS vehicle travels 200 miles per day and serves ten passengers. The AVO for vehicles is assumed to be 1.25. It is assumed that 1 SAV can serve about 200 PMT with an average party size of 1.25. The need for charging with PHEVs and BEVs is assumed to induce an additional 4.5% VMT (Chen et al., 2016). Similar to the AV scenario, a Bass diffusion model is used to interpolate DRS market share between the implementation year of 2025 and horizon year of 2050.

The GHG benefits of DRS come in two forms. First, the use of fleet vehicles provides greater control over the fuel stock, meaning the vehicle population could be more quickly transitioned towards PHEVs and BEVs than relying on market forces and individual consumer adoption. Second, the higher utilization of SAVs (i.e., each serving 5-10 travelers) may reduce the population of vehicles and associated embodied emissions relative to private vehicle ownership. However, the higher vehicle turnover (about every 2-3 years rather than 14-17 years) means that embodied emissions may be unchanged with DRS.

6.3. Precedents

According to the 2019 TBI, there are 11.5 thousand miles of micro-mobility travel and 357.3 thousand miles of ride-hail travel in MSP on a typical day (using weighted trip records). Ridehail and micro-mobility trips account for 0.24% and 0.03% of trips, respectively. For ride-hail trips, approximately 4.6% are reported as using a shared option (i.e., Lyft Line or Uberpool), although these trips may not necessarily be shared trips. Within St. Paul, the average micro-mobility trip distance is 3.1 miles among 34 reported trips. The average ride-hail trip distance is 13.9 miles among 271 trips reported in the TBI. Morgan Stanley predicts that shared rides will account for 26% of global travel by 2030, up from a 2015 share of 4% (Morgan Stanley, 2016).

6.4. Case Studies

There are no operational case studies of DRS, so the focus here is on simulations. A simulation study in Austin, TX found that SAVs produce 13.8% to 44.1% fewer emissions than human-driven vehicles due to their more efficient driving patterns (Fagnant and Kockelman, 2014). However, this result only holds if total VMT is unchanged. Empty VMT for the SAV fleet accounts for 7.8% to 14.2% of total VMT in the simulations.

6.5. Cost Estimation

Beyond the per mile costs outlined in section 1.6.5, the carbon abatement costs of DRS require additional analysis beyond the scope of this project.

6.6. Equity

DRS is anticipated to improve equity by providing a higher quality service to low-income households. However, it will be important for public agencies to work with DRS service providers to ensure an equitable service for all users, including the elderly and disabled populations.

6.7. Barriers, and Enablers

DRS faces a similar set of political, social, and technical challenges to AVs.

7. Road Pricing

Road pricing is likely to be an important component of transportation policy going forward. It provides a clear signal to travelers about the cost of driving and contains its own revenue stream. It is also a key component of AV and DRS policy because it helps to address the problem of induced VMT predicted to occur with the adoption of these technologies. Finally, road pricing provides an alternative revenue stream to the gas tax as vehicle efficiencies increase and EV adoption grows. The tool includes three alternatives for road pricing. The first is a simple VMT fee that charges for each mile of travel. The second is a pay-as-you-drive (PAYD) insurance plan. Finally, the tool includes a crude representation of decongestion pricing. Ideally, the congestion price should vary between the time of day and road link based on the current level of congestion. The simplified nature of the GHG tool makes such an implementation infeasible. A similar method to the CFB study (Porter et al., 2019) is followed, whereby an elevated VMT fee can be applied to a portion of travel approximating the percent of VMT that is congested.

7.1. GHG Impact Calculation

A change in VMT from the introduction of travel pricing can be introduced to the GHG impact equation for transportation as an elasticity factor.

A study by MAPC in Massachusetts used a combination of the VisionEval transportation model and UrbanSim land use model to examine land use and pricing scenarios (Harmon, 2021). The introduction of a 25 cent per mile VMT fee is anticipated to lower VMT by 15%. They find that even a tripling of the gas tax from 24 cents per gallon to 75 cents per gallon would only slow VMT growth by 1.5%. However, these results are highly dependent on the choice of elasticity and marginal cost. It is also important to note the variation in elasticity obtained in different jurisdictions. In one study, the long-run gasoline price elasticity in the United States is found to be -0.23 compared with -1.35 in Europe. This variation in the elasticity is attributable to the historically much higher fuel taxes in Europe leading to the adoption of more fuel-efficient vehicles and more robust transit networks. The implication is that alternative fuel vehicles, transit, and CUD are important complementary policies to pricing and will likely become more important as the population becomes more sensitive to marginal changes in the price of travel.

Spiller et al. (2014) estimated the price elasticity of VMT at -0.511, which is on the higher end of the range found in the literature. Their higher estimate is partially due to the consideration of multi-vehicle households switching to more fuel-efficient vehicles in response to increases in fuel price. In the case of a VMT fee, this option would not be available to these households.

In the CFB analysis, they assume that 29% of urban VMT is congested as an input to the determination of congesting pricing (Porter et al., 2019). Their estimate is in line with the findings of TAM for the Boston Metropolitan Area, which gives an estimate of 29.5%. However, this estimate is for peak VMT, which the TAM team defines as 6 AM to 10 AM and 3 PM to 7 PM. The same TAM study estimates peak VMT congestion in MSP as 24.6%. We assume there is no congestion during non-peak periods and estimate the percent of VMT during the peak period, as a proportion of total VMT for MSP, using weighted TBI 2019 results. PMT is converted to VMT using an AVO of 1.14 and 1.63 for commuting and non-commuting trips, respectively (based on USDOT (2010) statistics). The adjusted congested VMT is then 10.9% of all travel. This estimate was compared with an MNDOT (2019) estimate for highway VMT of 23.2% in 2017. We calculated a secondary set of congestion estimates using TBI data for each CTU. It is assumed that most congestion occurs in the peak periods on weekdays. TBI trips are selected that occur wholly within the same CTU and average travel times estimated for each of peak and off-peak periods. This method suggests an approximately 13% increase in average travel times during peak periods relative to off-peak periods.

7.2. Precedents

Minnesota examined mileage-based user fees (MBUF) through a pilot program focused on both technical feasibility and traveler attitudes (Rephlo, 2013). The study included 500 residents of Wright County, which is adjacent and to the west of the 7-county jurisdiction of Met Council. The MBUF rate was set at \$0.03 per mile for travel during peak periods or within an area bounded by I-494 and I-694 and \$0.01 per mile for all other travel. The pilot test generated \$40,000 of *simulated* revenue from an average monthly fee of \$20 per traveler. The cost was designed to be roughly commensurate with the Minnesota fuel tax. Only 17% of participants felt the MBUF was higher than they anticipated. Preference for an MBUF rather than a fuel tax was attributed to it providing a more equitable distribution of transportation costs. Those opposed to the change cited the clearer tax (i.e., monthly bill) versus the current system of incorporating the tax into the price of fuel at the pump. Data privacy was not a significant concern among participants, although there is some question as to whether self-selection bias may be a factor since participation in the study was voluntary. The pilot study also took advantage of the GPS technology used to track VMT to provide safety notifications to participants. These notifications included alerts when the driver exceeded the posted speed limit or entered a highway work zone. Many participants preferred that these elements be removed, and the focus placed on MBUF. While the project was motivated by a need to find alternative revenue streams due to increased vehicle fuel efficiency, the use of a *simulated* price means that there was no significant reduction in VMT among participants.

7.3. Case Studies

Road pricing systems have a longer history in Europe than North America. London implemented a cordon pricing system in 2003 for its CBD. The charge is currently set at £15 (US\$21) and an additional

£10 (US\$14) is levied on vehicles that do not meet the Euro 4 emissions standard. Similar systems are in place in Stockholm and Milan (Crocchi, 2016). It is estimated that the London and Stockholm systems have reduced daily CO₂ emissions by 13-16% (Anas and Lindsey, 2011; Transport for London, 2005). However, note that the VMT and GHG reductions associated with these charges are likely much higher than one would expect in MSP. The estimated **price elasticity** of VMT in the London and Stockholm case studies is in the range of -0.47 to -0.70 (Börjesson et al., 2012; Transport for London, 2008), whereas this elasticity is more likely -0.10 to -0.30 (Goodwin et al., 2004; Hymel and Small, 2015) in MSP due to the lower ability to switch modes.

California, Washington, and Oregon have piloted VMT fees. In California, users could choose from four monitoring systems: a time permit, a mileage permit, an odometer charge, or automated mileage reporting (Clements et al., 2020). The automated reporting option has the advantage that users are not charged for out-of-state or private road travel. Similarly, in Oregon users have the option of a GPS tracker or non-tracking options, such as odometer reading. The California pilot program uses simulated pricing while the Oregon pilot charges real prices.

7.4. Cost Estimation

The carbon abatement cost of road pricing is not reported in the literature. Recommendations for suitable VMT fees or PAYD insurance rates are taken from past studies. The CFB study used quite high prices of \$0.25 per mile for a VMT fee and \$0.50 for decongestion pricing (Porter et al., 2019). We suggest using these values as upper thresholds. In the MNDOT study referenced above (Rephlo, 2013), they use a \$0.03 per mile fee, which is a good lower limit on the fee. Cavallaro et al. (2018) summarized several studies of VMT fees and PAYD insurance. Their finding is that VMT fees in research studies vary between \$0.05 per mile to \$1.72 per mile (in 2015 and converting from euros per km to US\$ per mile). PAYD insurance was assumed to be \$0.12 in a U.S. study (Bordoff and Noel, 2008).

7.5. Equity

A common criticism of road pricing is that it disproportionately affects lower-income households who are less able to pay for the higher cost of travel. There are several ways to address the regressive nature of the policy. One such policy is credit-based decongestion pricing (CBCP). Under CBCP, travelers are allocated a monthly travel budget, providing for a minimum level of road usage for all. Larger credit allocations would be provided to travelers who chose to shift their departure time or mode. Credits can then be used at a later date when the traveler needs to travel during a peak period when the price is high. Those who choose to change their departure time or mode generally have a lower value of travel time (i.e., lower-income), so the distribution of credits tends to address the equity issue of road pricing. A simulation study in Austin, TX found an average credit to drivers between \$2 to \$65, depending on the details of its implementation (Li et al., 2020). Tolling the entire road network produces benefits (measured as travel time savings less toll cost) for 99% of residents.

7.6. Barriers and Enablers

Public support is a major barrier to implementing road pricing. One strategy for increasing support is to include citizen-led committees in the shaping of the policy. This approach would help to encourage buy-

in, increase transparency, and ensure an equitable and effective policy. There are also a variety of technical problems associated with fully dynamic road pricing. Prices should vary by location and time of day, and in the optimal case be variable across individual travels. Such a policy would require GPS transponders in vehicles, which could take time to ensure across the vehicle fleet. Charging a higher fixed rate for **non-transpondered** vehicles could help to speed up the rate of adoption (Clements et al., 2020).

8. Parking Policy

Parking is an important component of the transportation system. A lack of parking can result in additional VMT from vehicles searching for a parking spot (up to 30% of vehicles in downtown traffic (Shoup, 2006)). Surface lots and **parkades** take up valuable space that could be put to more useful purposes. The most common strategies for controlling parking demand are increasing its price and restricting the time that parking is available. These strategies have been reviewed in detail and price elasticities are available for the price of parking on VMT (see below). However, a strategy that is seeing increasing applications is addressing the underlying parking requirements in buildings. While promising, empirical evidence of the impact of such policies on transportation demand remains sparse.

The long-run **elasticity** of VMT to parking cost is estimated to be in the range of -0.18 to -0.45 based on a **meta-analysis** of values reported in the literature (Lehner and Peer, 2019). The 2019 TBI includes data on the cost of parking at trip destinations. However, in most cases, the reported cost is \$0. In the absence of a parking price from the TBI data, a minimal price of \$0.01 is assumed for CTUs.

The long-run **elasticity** of VMT to parking cost is estimated to be in the range of -0.18 to -0.45 based on a meta-analysis of values reported in the literature (Lehner and Peer, 2019). On-street parking in central St. Paul costs between \$0.50 to \$2.00 per hour. However, parking is free in most locations and at most times of the day. In the absence of better data, we use an estimate of the search cost found for Melbourne, Australia of \$1.00 per hour in St. Paul. For the other two CTU (Lake Elmo and Shoreview), it is assumed that the search cost is considerably lower and an estimate of \$0.10 per hour is used as the marginal parking cost. TBI data bear out this assumption, with average monetary parking costs of \$0 for trips ending in Lake Elmo and Shoreview and \$0.22 for trips ending in St. Paul.

8.2. GHG Impact Calculation

Similar to road pricing, a change in VMT from the introduction of parking pricing can be introduced to the GHG impact equation for transportation as an elasticity factor.

8.3. Precedents

On-street parking in central St. Paul costs between \$0.50 to \$2.00 per hour (SRF and Nelson Nygaard, 2015). However, parking is free in most locations and at most times of the day in the city. As of 2015, St. Paul already exempted all downtown land from minimum parking requirements.

8.4. Case Studies

Washington D.C. implemented variable parking prices in 2008 in an effort to maintain occupancy at, or below, 85% in key areas (US DOT, 2012). Data were collected using license plate recognition (LPR)

technology. In the Ballpark District, it was found that the rate was set too high initially - occupancy dropped to 34 percent on game days and 24 percent on non-game days. Seattle implemented variable pricing in 2010, with rates varying between \$1 to \$4 per hour (US DOT, 2012). The city was separated into 23 parking districts, with different pricing and time limit regimes.

8.5. Cost Estimation

A surface parking space costs about \$24,000 based on the cost of construction and maintenance (Shoup, 2014). The cost of an underground parking space in many urban markets is about \$30,000–\$50,000 (Williams et al., 2016).

8.6. Equity

In addition to the direct cost of parking, other costs arising from the oversupply of parking that affect equity. In San Francisco, it was found that single-family detached houses and condominiums were more than 10 percent more expensive if they included off-street parking (Jia and Wachs, 1999). By removing such requirements from land use and zoning ordinances, many more households would qualify for mortgages.

8.7. Barriers/Enablers

As with any policy that imposes a monetary cost on the user, public acceptance is a major implementation barrier to higher parking prices. [Ameliorating efforts](#) might include matching pricing to transit, such that higher prices are charged in areas with better access to alternative travel modes. Technology provides a means of making prices more dynamic so that high prices are not charged for parking during times when demand is low. The ability to measure occupancy in real-time and dynamically change the price of parking is increasingly possible and cost-effective (US DOT, 2012).

9. Telework

Telework is the concept of working from home to eliminate the need to commute to work. While it holds some promise for reducing GHG emissions, its effectiveness is less clear when one considers induced travel and additional energy demands placed on the home.

9.1. GHG Impact Calculation

A scenario factor (SF) can be introduced to the generic GHG impact equation as an adjustment based on the effect of telework on both transportation and building energy. At a first approximation, telework eliminates commute VMT. However, the net effect on VMT and household GHG production is less clear. Individuals who work from home tend to make additional non-commute trips during their additional discretionary time (Kim et al., 2015; Mokhtarian, 2009). Other household members also can use the now available vehicle for their commute (potentially switching away from transit) or for other trips. Finally, working from home introduces additional requirements on household energy consumption: more daytime heating and, potentially, the need for a larger dwelling to provide space for a home office. The most recent studies that also include energy associated with working from home find an effective

reduction of 2.5-2.8% in household CO₂ emissions per telework day (Fu et al., 2012; Larson and Zhao, 2017).

An important secondary question regarding telework is its market potential. It is estimated that about 35 percent of jobs in the United States can be performed from home full-time (Dingel and Neiman, 2020). Prior to the COVID-19 pandemic, time use data for the United States indicates that less than a quarter of workers worked from home, and most spent less than half their working hours at home. The portion of jobs that can be done from home varies by metropolitan area. In MSP, approximately 38 to 52 percent of jobs can be done fully through telework (Dingel and Neiman, 2020).

9.2. Precedents

The number of workers who teleworked at least one day per month grew in Minnesota from 18 percent in 1994 to 24 percent in 1997. As early as 1999, one-third of 200 surveyed businesses reporting allowing their employees to telework. Comparing a private firm and public agency implementation of telework in Minnesota, Wells et al. (Wells et al., 2001) found that the private firms implemented telework to recruit skilled workers while the public agency was responding to budget constraints. While the two employers had different motivations, they would receive similar benefits from attracting labor and reducing operational costs. It was also found that telecommuters were less likely to use transit for their commute. This difference is likely a result of the lower financial benefit from purchasing a monthly pass when not commuting to the office every day. To the best of our knowledge, such impacts of large-scale adoption of telework on public transit have not been examined in detail. Using the 2010 TBI, Yum found that telecommuters generally have more congested trips on the days they commute to the workplace than regular commuters (Yum, 2021). Telecommuting is more common among residents of central communities in MSP. This more recent finding suggests a shift in the characteristics of teleworkers from the early study by Wells et al. in 2001; however, the study contexts are quite different, and it is difficult to draw a strong conclusion.

An interesting analysis by the International Energy Agency found that, if everyone in the world who can work from home did so one day per week, it would reduce annual GHG emissions (transportation+residential) by 24.2 Mt CO₂e - roughly equivalent to the annual emissions in Greater London (or about 0.07% of global emissions) (Crow and Millot, 2020). While not a large magnitude reduction, this result indicates that telework should be considered in the suite of decarbonization strategies. Also, we do not have a clear precedent for the widespread adoption of telework. It may be the case that such a policy would encourage diversification of land uses in suburban communities as residents seek out the urban amenities they previously found near their workplace (e.g., restaurants and shopping opportunities near their home). These changes would potentially encourage additional VMT reductions through the benefits of the 5Ds of urban form.

9.3. Case Studies

Many companies have policies in place to support working from home. Among the earliest adopters of telework was AT&T (in 1992). As of 2000, AT&T workers were already avoiding 110 million annual miles of driving to the office (Papazian, 2002). In 2016, Dell found that their employees were working from home on average 9 days per month under their Connected Workplace program (Pflueger et al., 2016).

They estimate annual GHG savings of 40 kt CO₂e from reduced travel and energy demand in their offices.

Fergus Falls, MN advertises itself as the “Telework Capital of Minnesota” (Fergus Falls Telework Committee, 2014). The initiative began in 2008 as a partnership between the Fergus Falls Economic Improvement Commission and the local internet provider. They were interested in growing rural jobs following the onset of the 2008 recession and were inspired by the success of Perry County, TN with a similar initiative. They provide short-term office space in “telework hotels” for when teleworkers require a space for meetings, making photocopies, or confidentiality. The hotel is now part of a larger business development center, which offers additional conference spaces and low rent office space for startups.

9.4. Cost Estimation

There are no good literature values for the carbon abatement cost of telework. There is a potential to reduce office costs if space is allocated more strategically among employees.

9.5. Equity + Barriers and Enablers

The COVID-19 pandemic has provided a case study in telework on several fronts. There is a clear income difference in the ability to telework, with lower-wage jobs being significantly less amenable to it. Variations also exist by gender. Women take on a disproportionate share of the parenting task and have been more strongly affected over the past year (Zamarro and Prados, 2021). In the spring of 2020, unemployment among women was significantly higher than for men. While this difference has since disappeared, labor force participation (i.e., those actively seeking work) remains lower among women (Zarrilli and Luomaranta, 2021). The increasing prevalence of telework raises the need for improved support of parents through childcare subsidies and other measures.

10. Transportation and Land Use Behavioral Nudges Case Studies

Most applications of behavioral nudging in transportation provide the public personalized information to encourage them to more frequently use active modes (i.e., walking or cycling) or public transit. The key points of contact for transit users are registration of a transit (smart) pass, signup for bulk passes or auto-refill, use of a trip planning app, riding transit for commutes, and riding transit for non-commute trips. Similarly, there are opportunities to consider nudges when creating a new LRT or bus line, making changes to a transit stop, or increasing service frequency.

Durham, North Carolina residents provided their home and work addresses to the city through a phone app, which then provided them with maps and travel times for bike, bus, and walking routes (Gardner, 2019). Participants in the program were 12 percent more likely to use alternative modes of travel than those who did not receive the nudge. In a second study, city employees were entered into a draw with a cash prize when they chose to ride the bus over driving alone. Participants reported a 19 percent increase in alternative mode commuting. Bloomberg Philanthropies has since awarded the city \$1 million through their U.S. Mayors Challenge to continue their nudge campaign.

Metro Vancouver, in Canada, developed a detailed program of behavioral nudges to encourage the use of public transit (Alta Planning + Design, 2018). They started by identifying the key points when people can be nudged to change their travel choice, designed a set of interventions, and targeted the

interventions by user group. For low-frequency users, it was found that individuals were more amenable to nudge during times of transition (e.g., moving, getting married, having a child, or starting a new job). It was also found that these users had negative associations with transit. By addressing these biases, these users were more likely to consider transit. The second group identified was medium-frequency users, representing about 11 percent of the regional population. They were familiar and comfortable with the system but only used transit a few times each month. Such users often see their car ownership as a sunk cost, making them feel that they must use their car to justify its cost. One strategy the team explored was introducing **pre-loaded smart cards**, which introduce a similar sunk cost and may encourage more frequent transit use.

A study in Boston provided a 50% discounted MBTA fare to low-income individuals in a random control trial (Rosenblum et al., 2019). These individuals took 30% more transit trips than the control group and made more trips for health care and social services. These results highlight the social role of transit.

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